Questions

Q1.

(i) Find

$$\int_5^{13} \frac{1}{(2x-1)}\,dx$$

writing your answer in its simplest form.

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(4)

(ii) Use integration to find the exact value of

$$\int_0^{\frac{\pi}{2}} \sin 2x + \sec\frac{1}{3}x\tan\frac{1}{3}x\,dx$$

(3)

(Total for question = 7 marks)

(**Q04 WMA13/01, Specimen papers**)

Q2.

(i) Find

$$\int \frac{12}{(2x-1)^2}dx$$

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giving your answer in simplest form.

(2)

(ii) (a) Write $\frac{4x+3}{x+2}$ in the form

$A + \frac{B}{x+2}$ where A and B are constants to be found

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(b) Hence find, using algebraic integration, the exact value of

$$\int_{-8}^{-5} \frac{4x+3}{x+2} \mathrm{d}x$$

giving your answer in simplest form.

(6)

(Total for question = 8 marks)

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(Q03 WMA13/01, June 2021)

Q3.

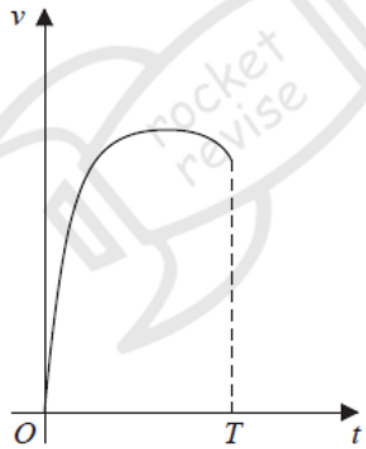


Figure 4

Figure 4 is a graph showing the velocity of a sprinter during a 100 m race.

The sprinter's velocity during the race, v m s^{-1}, is modelled by the equation

$$v = 12 - \mathrm{e}^{t-10} - 12\mathrm{e}^{-0.75t} \qquad t \geq 0$$

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where t seconds is the time after the sprinter begins to run.

According to the model,

(a) find, using calculus, the sprinter's maximum velocity during the race.

(5)

Given that the sprinter runs 100 m in T seconds, such that

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$\int_0^T v\,dt = 100$

(b) show that T is a solution of the equation

$$T = \frac{1}{12}(116 - 16e^{-0.75T} + e^{T-10} - e^{-10})$$

(4)

The iteration formula

$$T_{n+1} = \frac{1}{12}(116 - 16e^{-0.75T_n} + e^{T_n-10} - e^{-10})$$

is used to find an approximate value for T

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Using this iteration formula with $T_1 = 10$

(c) find, to 4 decimal places,

(i) the value of T_2

(ii) the time taken by the sprinter to run the race, according to the model.

(3)

(Total for question = 12 marks)

(Q08 WMA13/01, June 2022)

Q4.

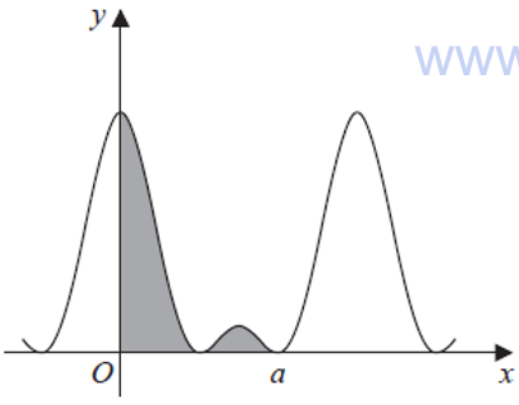


Figure 4

Figure 4 shows a sketch of part of the curve with equation

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$$y = (1 + 2\cos 2x)^2$$

(a) Show that

$$(1 + 2\cos 2x)^2 \equiv p + q\cos 2x + r\cos 4x$$

where p, q and r are constants to be found.

(2)

The curve touches the positive x-axis for the second time when $x = a$, as shown in Figure 4.

The regions bounded by the curve, the y-axis and the x-axis up to $x = a$ are shown shaded in Figure 4.

(b) Find, using algebraic integration and making your method clear, the exact total area of the shaded regions. Write your answer in simplest form.

(5)

(Total for question = 7 marks)

(**Q10 WMA13/01, Oct 2021**)

Q5.

Find

$$\int \frac{x^2 - 5}{2x^3} \mathrm{d}x \qquad x > 0$$

giving your answer in simplest form.

(3)

(Total for question = 3 marks)

(**Q01 WMA13/01, Jan 2021**)

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Q6.

Find

(i) $\int \frac{3x-2}{3x^2-4x+5}\,dx$

(2)

(ii) $\int \frac{e^{2x}}{(e^{2x}-1)^3}\,dx \qquad x \neq 0$

(2)

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(Total for question = 4 marks)

(Q09 WMA13/01, Jan 2021)

Q7.

In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

Given that k is a positive constant,

(a) find

$$\int \frac{9x}{3x^2+k}\,dx$$

(2)

Given also that

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$$\int_2^5 \frac{9x}{3x^2+k}\,dx = \ln 8$$

(b) find the value of k

(4)

(Total for question = 6 marks)

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(Q03 WMA13/01, June 2022)

Q8.

(i) Find, using algebraic integration, the exact value of

$$\int_3^{42} \frac{2}{3x-1}\,\mathrm{d}x$$

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giving your answer in simplest form.

(4)

(ii)

$$\mathrm{h}(x) = \frac{2x^3 - 7x^2 + 8x + 1}{(x-1)^2} \qquad x > 1$$

Given $\mathrm{h}(x) = Ax + B + \frac{C}{(x-1)^2}$ where A, B and C are constants to be found, find

$$\int \mathrm{h}(x)\,\mathrm{d}x$$

(6)

(Total for question = 10 marks)

(Q08 WMA13/01, Jan 2020)

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Q9.

(a) Given that

$$\frac{x^4 - x^3 - 10x^2 + 3x - 9}{x^2 - x - 12} \equiv x^2 + P + \frac{Q}{x-4} \qquad x > -3$$

find the value of the constant P and show that $Q = 5$

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The curve C has equation $y = g(x)$, where

$$g(x) = \frac{x^4 - x^3 - 10x^2 + 3x - 9}{x^2 - x - 12} \qquad -3 < x < 3.5 \qquad x \in \mathbb{R}$$

(b) Find the equation of the tangent to C at the point where $x = 2$
Give your answer in the form $y = mx + c$, where m and c are constants to be found.

(5)

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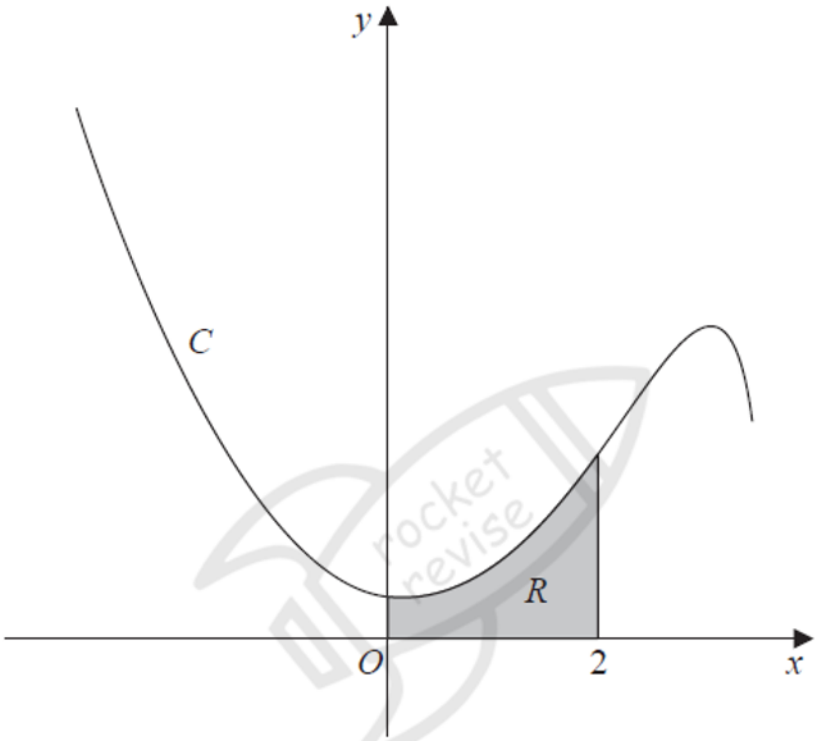


Figure 4

Figure 4 shows a sketch of the curve C.
The region R, shown shaded in Figure 4, is bounded by C, the y-axis, the x-axis and the line with equation $x = 2$

(c) Find the exact area of R, writing your answer in the form $a + b\ln 2$, where a and b are constants to be found.

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(5)

(Total for question = 14 marks)

(Q09 WMA13/01, Oct 2020)

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Q10.

(a) Show that

$$\sin 3x \equiv 3\sin x - 4\sin^3 x$$

(4)

(b) Hence find, using algebraic integration,

$$\int_0^{\frac{\pi}{3}} \sin^3 x \, dx$$

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(4)

(Total for question = 8 marks)

(Q05 WMA13/01, Oct 2020)

Q11.

In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

(a) Show that

$$\frac{\cos 2x}{\sin x} + \frac{\sin 2x}{\cos x} \equiv \operatorname{cosec} x \qquad x \neq \frac{n\pi}{2} \qquad n \in \mathbb{Z}$$

(3)

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(b) Hence solve, for $0 < \theta < \frac{\pi}{2}$

$$\left(\frac{\cos 2\theta}{\sin \theta} + \frac{\sin 2\theta}{\cos \theta}\right)^2 = 6\cot\theta - 4$$

giving your answers to 3 significant figures as appropriate.

(5)

(c) Using the result from part (a), or otherwise, find the exact value of

$$\int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \left(\frac{\cos 2x}{\sin x} + \frac{\sin 2x}{\cos x} \right) \cot x \, dx$$

(2)

(Total for question = 10 marks)

(Q09 WMA13/01, June 2023)

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